

Rice husk-derived mesoporous carbons via pore-tailoring as high-performance electrode materials of zinc-ion hybrid supercapacitors

Author links open overlay panelDawei Feng, Yuming Wang, Jiahao Li, Yiwen Liu, Bohan Guo

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Highlights

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Mesoporous carbon has a high mesopore ratio and a pore size distribution that matches well with hydrated zinc ions.

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Pore-tailoring of mesoporous carbon was achieved by regulating the silica self-template and the activation temperature.

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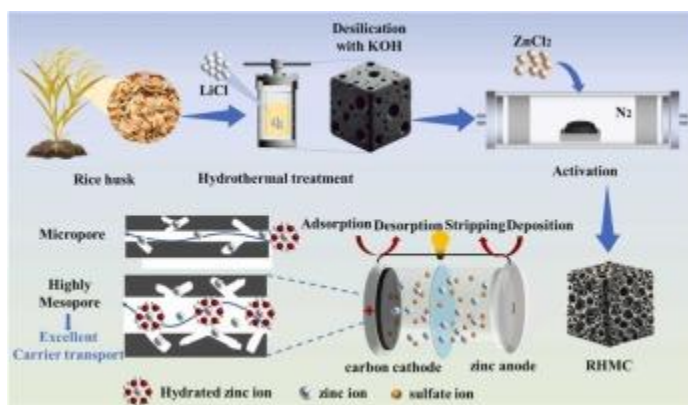
The aqueous zinc-ion supercapacitor shows a capacitance retention rate of 97.7% after 12000 charge-discharge cycles at 5 A g⁻¹.

Abstract

Biomass-derived carbon is a promising [electrode material](#) for aqueous zinc-ion [hybrid supercapacitors](#) (ZHSCs) due to its eco-friendliness and high-efficiency in energy storage. Previous studies mainly focus on the enhancement of specific [surface area](#); however, the optimization of [pore structure](#) is also important for energy storage performance of biomass-based [carbon materials](#), which is always ignored. Herein, a combination method of [hydrothermal treatment](#) and high-temperature activation is reported to synthesize rice husk-derived carbon, used as a high-performance cathode for aqueous ZHSC. Relying on natural SiO₂ template and regulation of activation temperature, [pore structure](#) of rice husk-derived carbon got effectively tailored, exhibiting a honeycomb-like structure with abundant [mesopores](#) and large micropores. The ZHSC with the rice husk-derived [mesoporous](#) carbon as the cathode exhibits both high specific capacitance and [energy density](#), achieving 172.5 F g⁻¹ at a [current density](#) of 0.5 A g⁻¹ and 77.61 W h

kg^{-1} at a power density of 449.8 W kg^{-1} . Additionally, the assembled ZHSC shows a capacitance retention rate of 97.7 % after 12,000 charge-discharge cycles at 5 A g^{-1} . This work successfully developed a rice husk-derived **mesoporous** carbon with pore-tailoring strategy, which offers an effective approach for transforming waste biomass into high-performance ZHSC electrode materials.

Graphical Abstract



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Introduction

To achieve sustainable development and environmental protection, the high-value utilization of waste biomass resources has become a hotspot in current research [1]. The conversion of biomass resources into efficient energy storage materials shows considerable promise in the renewable energy power system, which requires a large amount of materials and devices to ensure energy storage capacity and operational stability [2], [3].

In recent years, hybrid energy storage devices have become a research hotspot in energy storage field [4], [5], [6], [7]. Particularly, zinc-ion hybrid supercapacitors (ZHSCs) are expected to become promising electrochemical energy storage devices, which have advantages of combining high energy density from zinc-ion batteries and high power density as well as long cycle life from supercapacitors [8], [9], [10]. The cathode material is a critical factor determining the performance of zinc-ion hybrid supercapacitors [11]. Due to the low cost, environmental friendliness and excellent conductivity, biomass-derived carbon is considered one of the best cathode materials for ZHSCs [12], [13]. To enhance the capacitance of ZHSCs, most research efforts are dedicated to exploring biomass-derived carbon materials with higher specific surface areas, which usually relies on more abundant micropores. However, excessively small pores will lead to the micropore

confinement effect and hinder the storage of charge carriers, especially for aqueous ZHSCs [14]. As the primary charge carrier for ZHSCs during charge/discharge processes, $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$ is difficult to access the deep micropores of carbon materials due to its large ion size (0.86 nm), resulting in a severe loss of space utilization and poor Zn^{2+} storage capability [15], [16]. Therefore, pore structure tailoring is also important for preparing higher-performance biomass-derived carbon electrode materials for ZHSCs, which is always ignored.

Among the common agricultural biomass wastes, rice husks have an annual global production exceeding 100 million tons [17]. In recent years, rice husk-derived carbon materials have shown significant potential in the field of energy storage [18]. Compared with other biomass materials, rice husks possess a unique SiO_2 composition, which can serve as a natural template for tailoring the pore structure of derived carbon [19]. This holds promise for developing electrode materials that are more suitable for ZHSCs, while avoiding the high costs and complex procedures of traditional methods that regulate pore structure with template agents [20].

In this work, a rice husk-derived mesoporous carbon material used as high-performance aqueous ZHSC cathode was synthesized through a combination method of hydrothermal treatment and high-temperature activation. Unlike the direct carbonization methods commonly used in preparation of derived carbon [21], the procedure presented in this study achieves better regulation of the pore structure. Hydrothermal treatment weakens the biomass structure of rice husks and forms a significant distribution of macropores and mesopores. By further regulating the natural SiO_2 template and the activation temperature, the rice husk-derived carbon material exhibits a well-developed large microporous and mesoporous structure, providing abundant active sites and efficient transport pathways for zinc ions. This study shows that the rice husk-derived mesoporous carbon based on a pore-tailoring strategy exhibits outstanding performance in aqueous ZHSCs, which offers important insights for the preparation of high-performance biomass-derived electrode materials.

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Section snippets

Materials

The rice husks were sourced from Wuqing, Tianjin, China. All chemical used in the experiments, including LiCl , KOH , and ZnCl_2 , were analytical reagent (AR) grade and purchased from Kemai (Tianjin) Chemical Technology Co., Ltd.

Preparation of hydrothermal carbon

Firstly, the rice husks were crushed, sieved, and washed with deionized water, followed by drying at 80°C. The obtained rice husk powder (3 g) was mixed with LiCl (3 g) in 20 mL deionized water under ultrasonic stirring until a homogeneous mixture was achieved. The mixture

Characterizations of materials

The synthesis procedure of rice husk-derived mesoporous carbon (RHMC) was illustrated in Fig. 1. Under hydrothermal treatment catalyzed by lithium chloride, the cellulose, hemicellulose, and lignin components of rice husks undergo a series of reactions including dehydration, condensation, polymerization and aromatization, ultimately forming silica-embedded hydrothermal carbon. Hereafter, carbon materials with richer pores were obtained by high-temperature activation with ZnCl_2 , during which

Conclusion

This work presents a pore-tailoring strategy to synthesize rice husks-derived highly mesoporous carbon for use as a high-performance cathode material for ZHSCs. Benefiting from the highly mesoporous structure that matches the size of charge carriers, the prepared RHMC-700 showed outstanding carrier transport capability and reaction kinetics in aqueous ZHSCs, achieving a high specific capacitance of 172.5 F g^{-1} at a current density of 0.5 A g^{-1} and an energy density of $77.61 \text{ W h kg}^{-1}$ at a power

CRedit authorship contribution statement

Dawei Feng: Writing – review & editing, Resources, Methodology, Investigation, Conceptualization. **Yuming Wang:** Writing – review & editing, Resources, Methodology, Conceptualization. **Jiahao Li:** Writing – review & editing. **Yiwen Liu:** Writing – review & editing. **Bohan Guo:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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